Mathematics GCE O Level 2023 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (June 2023, Ordinary Level, 0570 Mathematics 1, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; mark only one answer for each question. Non-programmable calculators are allowed.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. The value of the digit 7 in 64.176 is

a) $\frac{7}{10}$
b) $\frac{7}{100}$
c) 7
d) 70

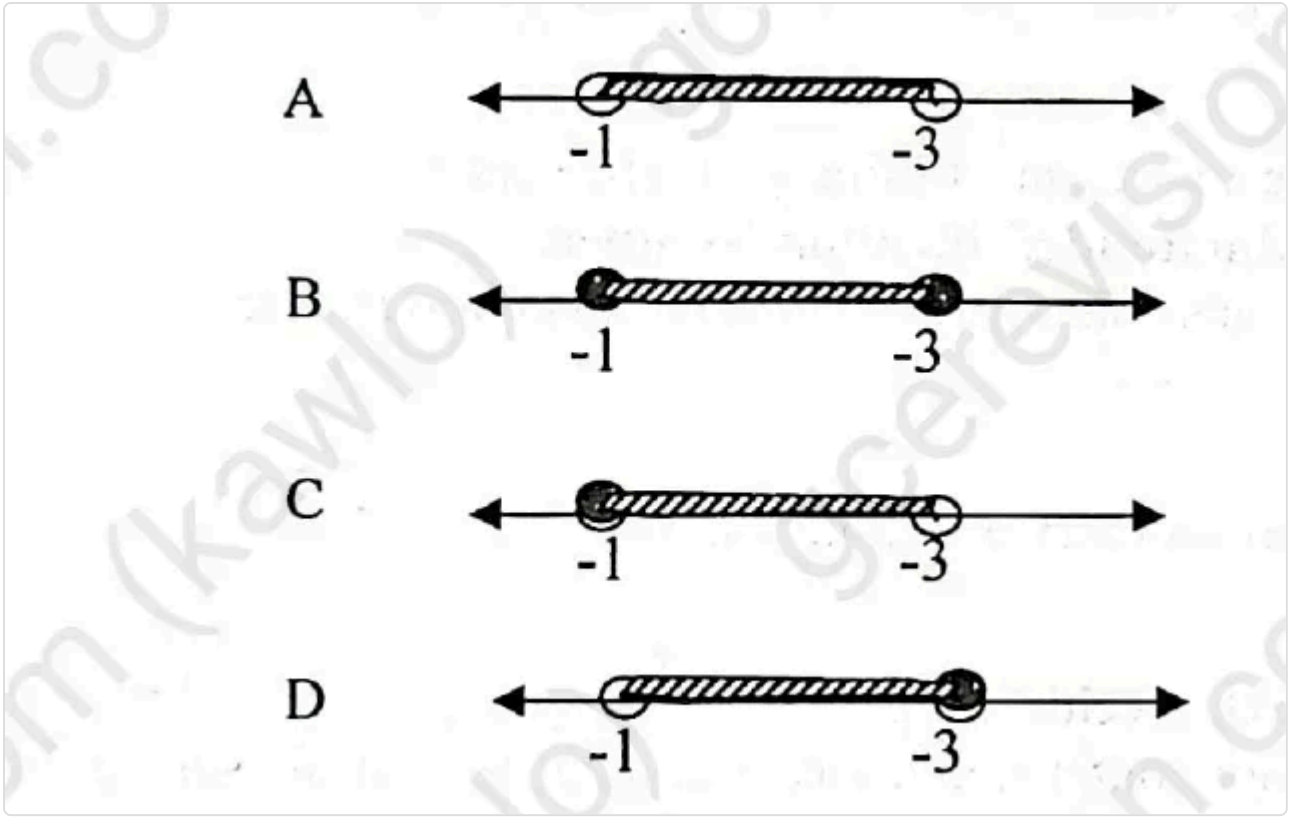
2. The H.C.F of $2^3 \times 3$ and $2 \times 3^2 \times 5$ is

a) $2 \times 3 \times 5$
b) $2^3 \times 3^2$
c) 2×3
d) $2^4 \times 3^2 \times 5$

3. The numbers $0.5,\ 0.4,\ 0,\ 0.25$ arranged in ascending order gives

a) $0,\ 0.25,\ 0.4,\ 0.5$
b) $0.25,\ 0.5,\ 0.4,\ 0$
c) $0.5,\ 0.4,\ 0.25,\ 0$
d) $0.25,\ 0.4,\ 0,\ 0.5$

4. The number line that represents the set $Q=\{-1



a) A
b) B
c) C
d) D

5. The number, $\sqrt{7}$, belongs to the set of

a) Integers
b) Natural Numbers
c) Rational Numbers
d) Irrational Numbers

6. The fraction *"four-thirds"* in figures is

a) $\frac{4}{3}$
b) $4\frac{1}{3}$
c) 4^3
d) $\frac{4}{1/3}$

7. Given that 0.00078 can be written as $a \times 10^n$, where $1

a) -3
b) 3
c) -4
d) 4

8. The number, 1949.46 to the nearest whole number is

a) 1949
b) 1950

c) 1940
d) 1900

9. In a class of 90 students, 25 are boys. The ratio of girls to boys is

a) $5 : 18$
b) $18 : 5$
c) $13 : 5$
d) $5 : 13$

10. The simple interest on $200{,}000$ FCFA at a rate of 3% after 5 years is

a) $30{,}000$ FCFA
b) $10{,}000$ FCFA
c) $15{,}000$ FCFA
d) $20{,}000$ FCFA

11. Given the sets $P = \{1, 3, 5, 6\}$ and $Q = \{2, 4, 7, 8\}$, then $n(P \cap Q)$ is

a) $\{0\}$
b) 0
c) $\varnothing$
d) $\{\varnothing\}$

12. Given the sets $F = \{\text{football players}\}$ and $G = \{\text{girls}\}$, the set notation for the statement *"All girls play football"* is

a) $G \cap F$
b) $G \cup F$
c) $G \subset F$
d) $F \subset G$

13. In set notation, the shaded region in figure 1 is

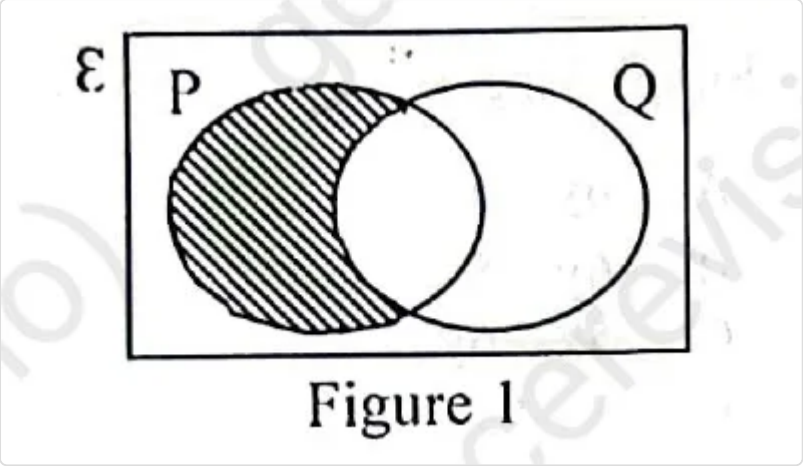


Figure 1

a) $P \cap Q'$
b) $P' \cap Q$
c) $Q' \cup P$
d) $P \cap Q$

14. Given that p: Adamu is hard working, q: Adamu is brilliant. The compound statement *"Adamu is hard working and brilliant"* is

a) an implication
b) a conjunction
c) a disjunction
d) a proposition

15. For any function f, f^{-1} always maps its

a) image to the range
b) range to its domain
c) domain to its range
d) domain to its codomain

16. Given that $f : x \mapsto 2x + 5$, then $f(-3)$ gives

a) -3
b) 1
c) -1
d) 3

17. Given that $f(x) = x$ and $g(x) = 3x - 2$, then the composite function $gf(x)$ is

a) $3x^2 - 2x$
b) $\frac{x-2}{3}$
c) $3x - 2$
d) $\frac{x+2}{3}$

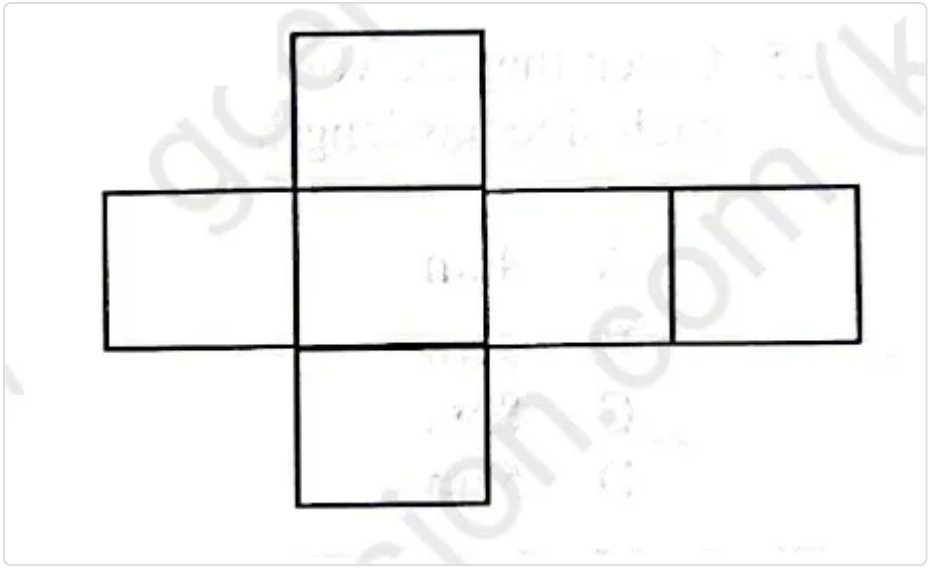
18. The distance from the centre of a circle to any point on the circumference is called

a) radius
b) an arc
c) a chord
d) a diameter

19. Given that the exterior angle of a regular polygon is 30°, then its interior angle is

a) 30°
b) 60°
c) 12°
d) 150°

20. The regular solid figure represented by the net in figure 2 is called



a) cube
b) cuboid
c) cylinder
d) square

21. The constructions shown in figure 3 are such that PQ is the perpendicular bisector of RS. The value of the angle PTS is

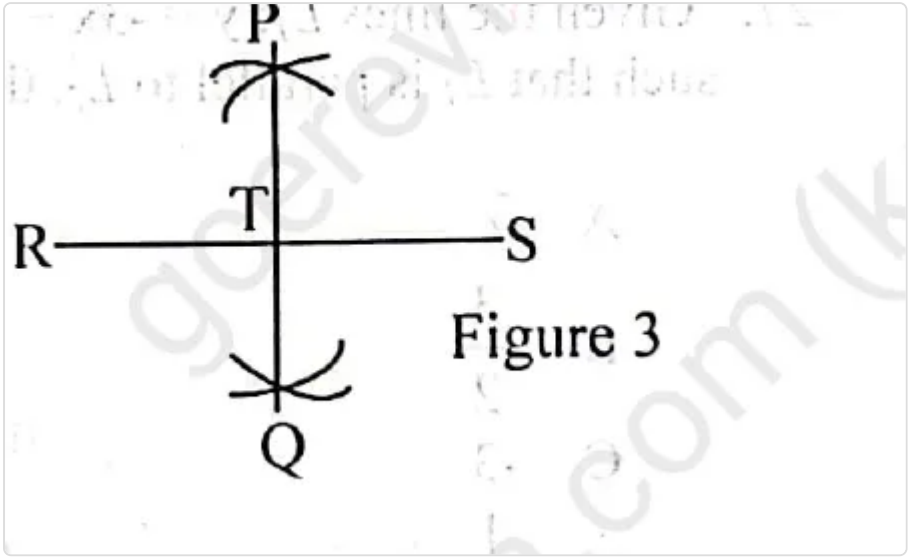


Figure 3

a) 180°
b) 60°
c) 45°
d) 90°

22. In figure 4, the value of angle p is

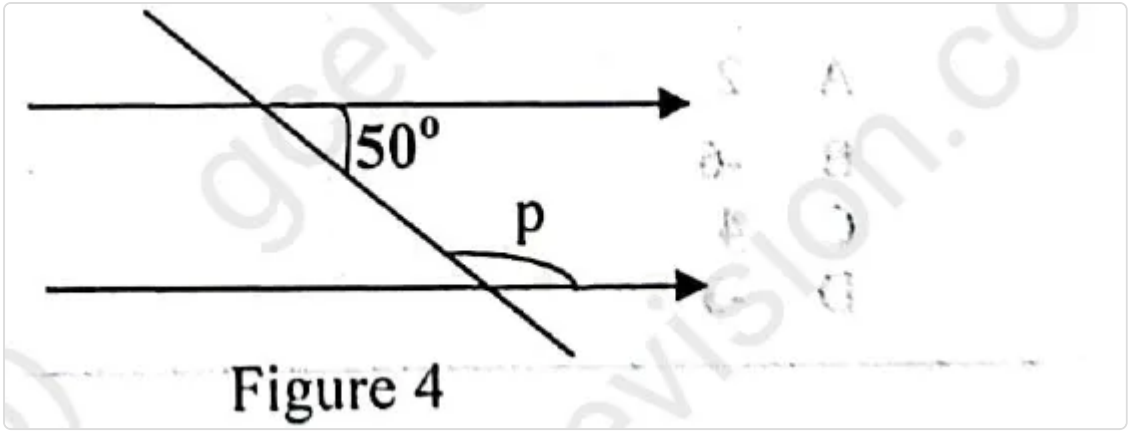


Figure 4

a) 160°
b) 50°
c) 130°
d) 40°

23. The perimeter of a rectangle is 24 cm. Given its width is 5 cm, then its length is

a) 8 cm
b) 7 cm
c) 5 cm
d) 4 cm

24. Given that $\pi = \frac{22}{7}$ and the radius of a circle is 7 cm, then the area of the circle is

a) $44\ \text{cm}^2$
b) $154\ \text{cm}^2$
c) $308\ \text{cm}^2$
d) $22\ \text{cm}^2$

25. Given that the volume of a cube is $64\ \text{cm}^3$, then each side has length

a) 4 cm
b) 3 cm
c) 8 cm
d) 5 cm

26. On the Cartesian plane, the line $y = 0$ is called the

a) Origin
b) x-axis
c) Ordinate
d) y-axis

27. Given the lines $L_1 : y = -3x + 3$ and $L_2 : y = mx - 2$, such that L_1 is parallel to L_2, the value of m is

a) 3
b) $\frac{3}{2}$
c) -3
d) $\frac{1}{3}$

28. The x-coordinate of the mid-point $M(x, y)$ of the points $P(0, 3)$ and $Q(4, -3)$ is

a) 2
b) -6
c) 4
d) -3

29. The number of terms in the expression $2ax + 3by - 1$ is

a) 7
b) 4
c) 2
d) 3

30. Simplifying $2^5 \times 2^{-3}$ gives

a) 2^{-15}
b) 2^{-2}
c) 2^2
d) 2^{-8}

31. Simplifying $3x - 6y - 2x$ gives

a) $6y - x$
b) $6y - 5x$
c) $5x - 6y$
d) $x - 6y$

32. Given that $P = \frac{E}{WL}$, an expression for W in terms of P, E and L is

a) $\frac{PL}{E}$
b) $\frac{E}{PL}$
c) PLE
d) $\frac{PE}{L}$

33. The next term in the sequence $3,\ 5,\ 8,\ 12,\ \ldots$ is

a) 17
b) 14
c) 16
d) 19

34. The range of values of x for which $7 + 2x < 11$ is

a) $x \leq 2$
b) $x < 2$
c) $x < 9$
d) $x \leq 9$

35. Given that the speed, V, of a car is inversely proportional to the time, t, then the equation relating V to t, where k is a constant is

a) $Vk = t$
b) $V = kt$

c) $V = \frac{k}{t}$
d) $Vk = \frac{1}{t}$

36. The number of regions in the network in figure 5 is

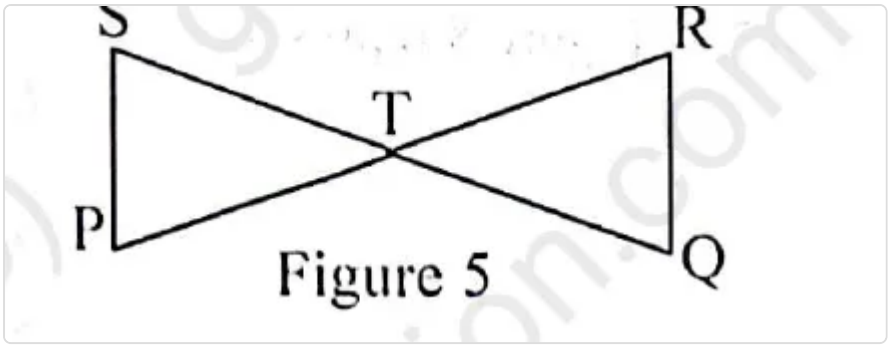


Figure 5

a) 6
b) 2
c) 5
d) 3

37. The longest side in a right-angled triangle is

a) Opposite
b) Adjacent
c) Hypotenuse
d) Acute

38. Given that $\sin A = \frac{2}{3}$. The cosine of the angle complementary to angle A is

a) $\frac{2}{3}$
b) $\frac{1}{3}$
c) 1.0
d) 1.5

39. The angle of elevation of Q from P, in figure 6 is

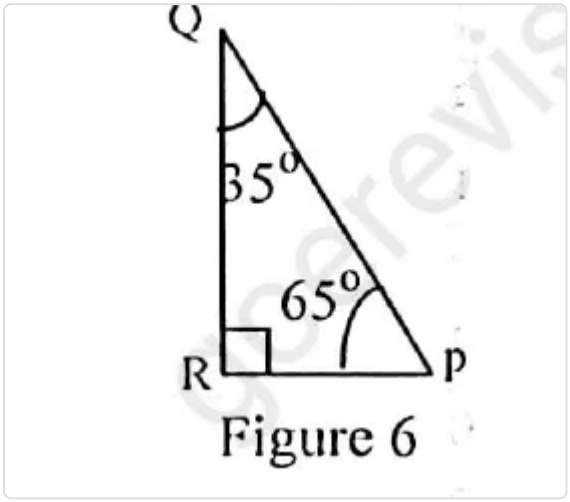


Figure 6

a) $65°$
b) $35°$
c) $90°$
d) $30°$

40. Given the vectors $\mathbf{p} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$, then $\mathbf{p} - \mathbf{r}$ is

a) $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$
b) $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$
c) $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$
d) $\begin{pmatrix} 4 \\ -4 \end{pmatrix}$

41. Given the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ in figure 7, then $\overrightarrow{PR}$ is

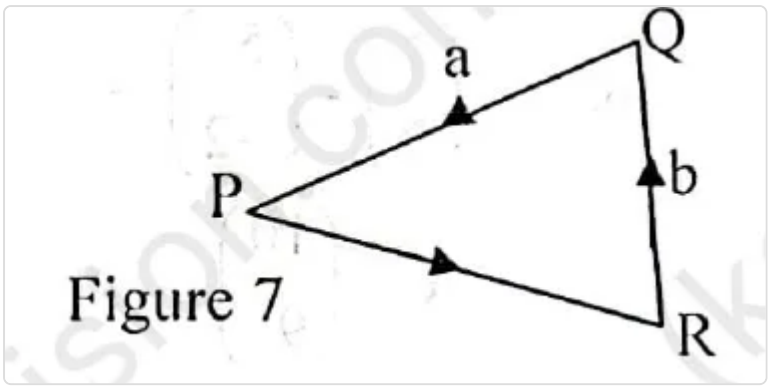


Figure 7

a) $a + b$
b) $a - b$
c) $-a + b$
d) $-a - b$

42. Given the vector $\overrightarrow{OM} = 2\mathbf{i} + 3\mathbf{j}$, the direction of this vector is

a) $\tan^{-1}\left(\frac{2}{3}\right)$
b) $\tan^{-1}\left(\frac{3}{2}\right)$
c) $\tan\left(\frac{3}{2}\right)$
d) $\tan^{-1} 5$

43. The leading diagonal elements of the matrix P, where $P = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$ are

a) 2 and 5
b) 3 and 4
c) 2 and 4
d) 3 and 5

44. The determinant of the matrix $\begin{pmatrix} 4 & 5 \\ 1 & 2 \end{pmatrix}$ is

a) -3
b) 3
c) 13
d) -13

45. The translation $\begin{pmatrix} 5 \\ -7 \end{pmatrix}$ maps the y-coordinate of the point $(2, 5)$ to

a) 2
b) -2
c) 12
d) -12

46. $\begin{pmatrix} 5 \\ 7 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \end{pmatrix} =$

a) $\begin{pmatrix} 10 \\ 9 \end{pmatrix}$
b) $\begin{pmatrix} 0 \\ 9 \end{pmatrix}$
c) $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$
d) $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

47. Figure 8 represents

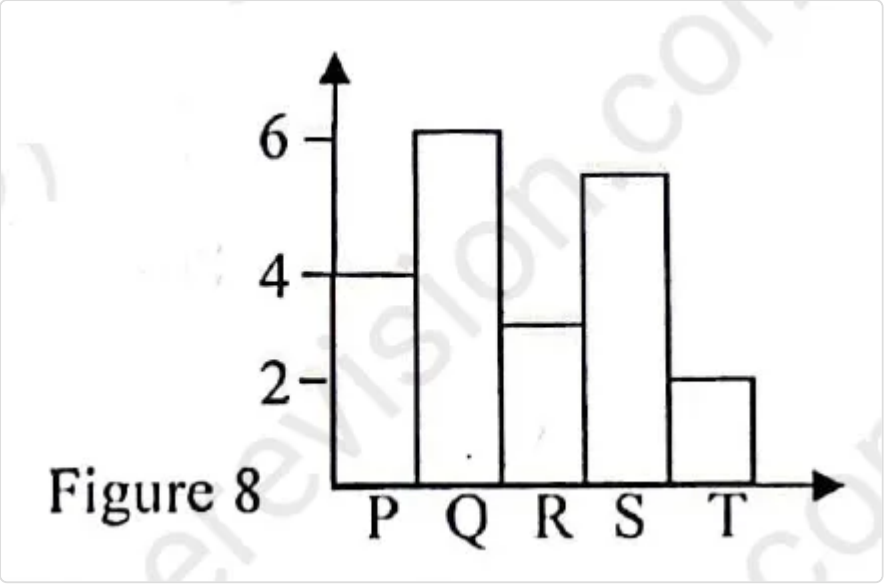


Figure 8

a) Frequency
b) Pie Chart
c) Bar Chart
d) Pictogram

48. The most popular mark scored in a test is called

a) Median
b) Mean
c) Data
d) Mode

49. The median of the scores $3,\ 4,\ 1,\ 2,\ 2$ is

a) 3
b) 4

c) 1
d) 2

50. The probability that Mary passes a test is $\frac{2}{3}$. The probability that she fails the test is

a) $\frac{1}{3}$
b) $\frac{1}{9}$
c) $\frac{4}{9}$
d) $\frac{2}{3}$

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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